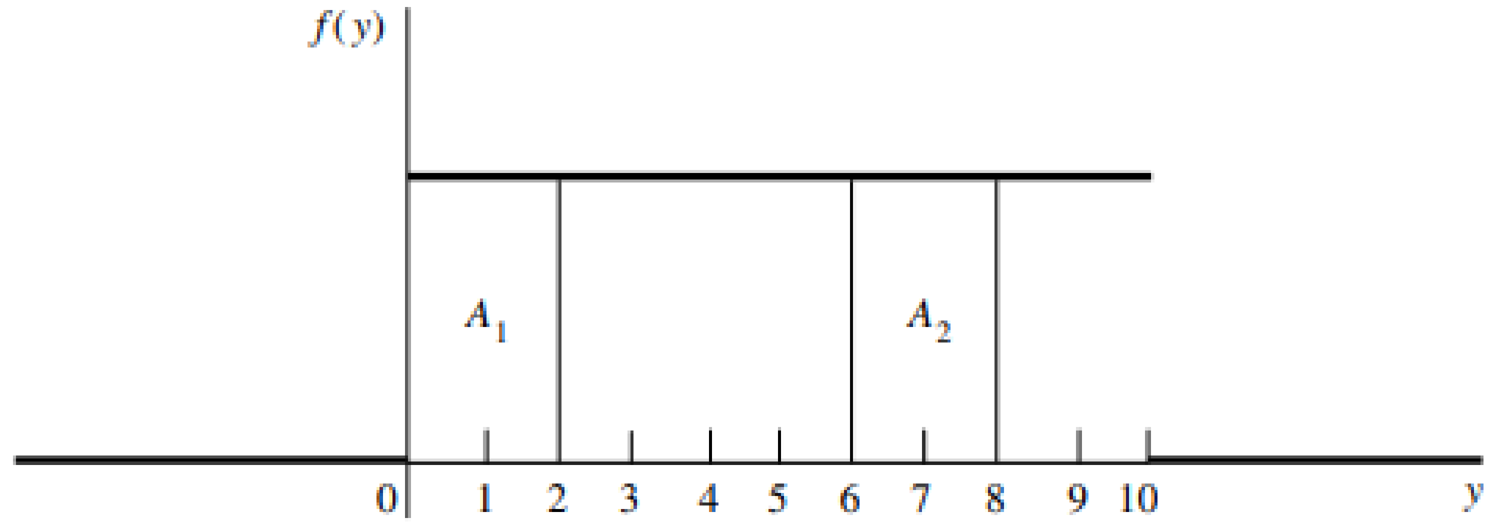
Special continuous random variables

CHAPTER 4

The Uniform Random Variable



f(y)
A1
A2
0
1
2
3
4
5
6
7
8
9
10
y

Definition

If $\theta_1 < \theta_2$, a random variable Y is said to have a continuous uniform probability distribution on the interval (θ_1, θ_2) if and only if the density function of Y is

$$f(y) = \begin{cases} \dfrac{1}{\theta_2 - \theta_1}, & \theta_1 \le y \le \theta_2, \\ 0, & \text{elsewhere.} \end{cases}$$

"what does the Uniform random variable model and why?"

The (discrete) Uniform Random Variable models the outcome of an experiment where the experiment is choosing an integer from $a, a+1, a+2, \ldots$, b at random.

Definition

The constants that determine the specific form of a density function are called parameters of the density function.

Example

Arrivals of customers at a checkout counter follow a Poisson distribution. It is known that, during a given 30-minute period, one customer arrived at the counter. Find the probability that the customer arrived during the last 5 minutes of the 30-minute period.

Theorem

If θ1 < θ2 and Y is a random variable uniformly distributed on the interval (θ1, θ2), then

$$\mu = E(Y) = \frac{\theta_1 + \theta_2}{2} \quad \text{and} \quad \sigma^2 = V(Y) = \frac{(\theta_2 - \theta_1)^2}{12}.$$

Example

The change in depth of a river from one day to the next, measured (in feet) at a specific location, is a random variable Y with the following density function:

$$f(y) = \begin{cases} k, & -2 \le y \le 2 \\ 0, & \text{elsewhere.} \end{cases}$$

1. Determine the value of k.
2. Obtain the distribution function for Y .

Example

Upon studying low bids for shipping contracts, a microcomputer manufacturing company finds that intrastate contracts have low bids that are uniformly distributed between 2 0 and 2 5, in units of thousands of dollars. Find the probability that the low bid on the next intrastate shipping contract

a is below $2 2 ,000.

b is in excess of $2 4,000.

The Exponential Random Variable

Definition

A random variable Y is said to have an exponential distribution with parameter β > 0 if and only if the density function of Y is

$$f(y) = \begin{cases} \frac{1}{\beta} e^{-y/\beta}, & 0 \le y < \infty, \\ 0, & \text{elsewhere.} \end{cases}$$

Theorem

If Y is an exponential random variable with parameter β, then $\mu = E(Y) = \beta$ and $\sigma^2 = V(Y) = \beta^2$.

The Standard Normal Random Variable

Introduction

- the position and shape of the normal curve depend on the parameters μ and σ
- the area below the normal curve and hence the probability for a given interval would vary for each different value of μ and σ
- Integrating the equation for the normal distribution each time a value is required can become tedious.
- This problem can be quickly rectified by introducing a standard normal distribution where $\mu = 0$ and $\sigma = 1$.

Definition

- *A* **standard normal random variable** *is a normally distributed random variable with mean μ =* 0 *and standard deviation σ* = 1. *It will always be denoted by the letter Z*.

The equation of the normal distribution curve

$$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

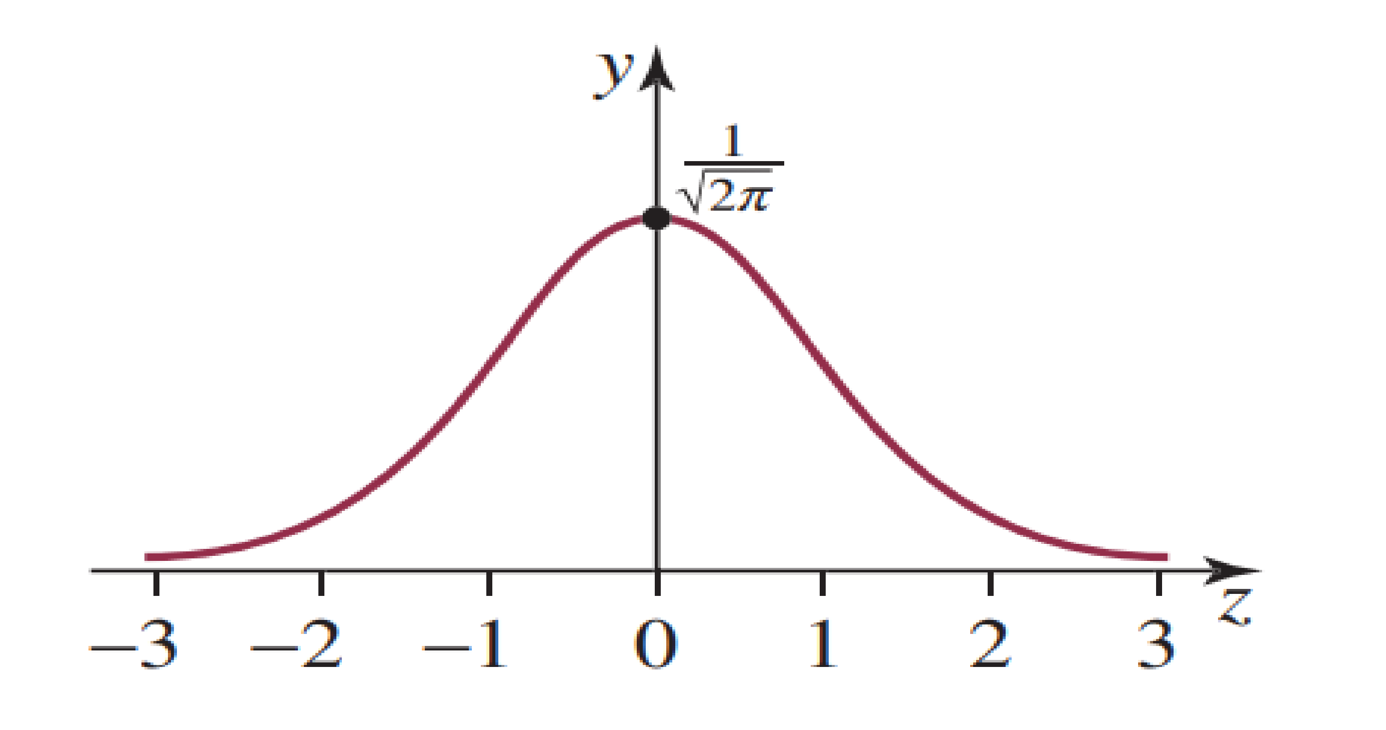
when converted to the standard normal curve becomes

$$g(z)=\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}$$

where $z = (x - \mu)/\sigma$ and $g(z) = \sigma f(x)$

The standard normal distribution is written as $Z \sim N(0, 1^2)$.

Density Curve for a Standard Normal Random Variable



Example

- Find the probabilities indicated, where as always Z denotes a standard normal random variable.
  I. $P(Z < 1.48)$.
  II. $P(Z < -0.25)$.

  III. $P(Z > 1.60)$.
  IV. $P(Z > -1.02)$

  V. $P(0.5<Z<1.57)$
  VI. $P(-2.55<Z<0.09)$

  VII. $P(1.13<Z<4.16)$
  VIII. $P(-5.22<Z<2.15)$

  IX. $P(-1<Z<1)$
  X. $P(-2<Z<2)$
  XI. $P(-3<Z<3)$

Probability Computations for General Normal Random Variables

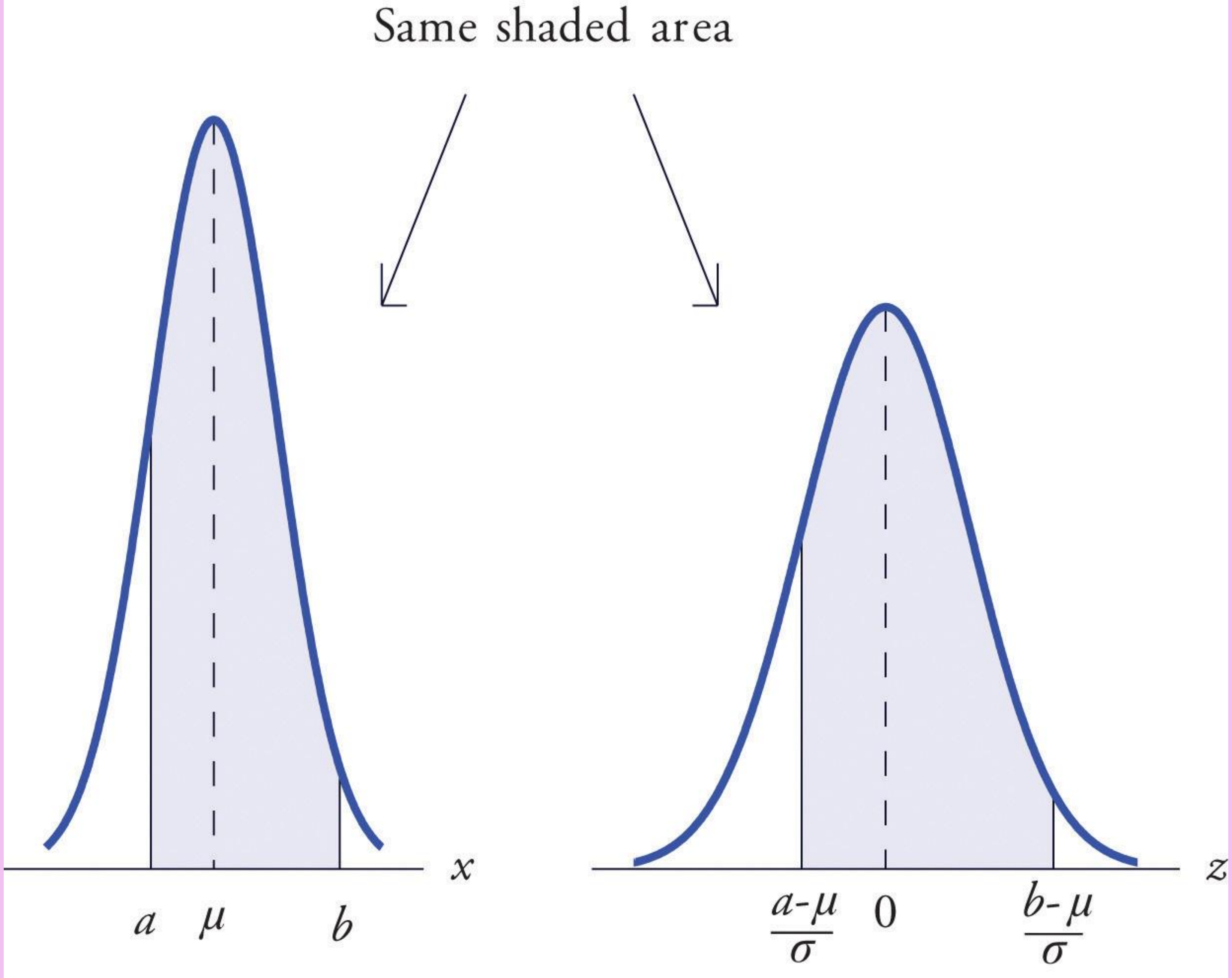
Introduction

- If X is a normally distributed random variable with mean μ and standard deviation σ, then

$$P(a < X < b) = P\left(\frac{a-\mu}{\sigma} < Z < \frac{b-\mu}{\sigma}\right)$$

- where Z denotes a standard normal random variable. a can be any decimal number or $-\infty$; b can be any decimal number or ∞.
- The new endpoints $(a-\mu)/\sigma$ and $(b-\mu)/\sigma$ are the z-scores of a and b

Probability for an Interval of Finite Length



Example

- Let X be a normal random variable with mean $\mu = 10$ and standard deviation $\sigma = 2.5$. Compute the following probabilities.

I. $P(X < 14)$

II. $P(8<X<14)$

- The lifetimes of the tread of a certain automobile tire are normally distributed with mean 37,500 miles and standard deviation 4,500 miles. Find the probability that the tread life of a randomly selected tire will be between 30,000 and 40,000 miles.

Chi-Square Random Variable

- Suppose X $_1$, X $_2$, · · · , X$_n$,
- are independent standard normal random variables.
- Then

$$\chi^2_{\mathbf{n}} \equiv X_1^2 + X_2^2 + \cdots + X_n^2 ,$$

- is called the chi-square random variable with n degrees of freedom.

$$E[\chi^2_n] = n \quad , \quad Var(\chi^2_n) = 2n \quad , \quad \sigma(\chi^2_n) = \sqrt{2n} .$$

- The 2 in χ 2 n is part of its name , while 2 in X_1^2 , etc. is “power 2 ” !

THE CENTRAL LIMIT THEOREM

Introduction

- The **central limit theorem (CLT)** is one of the most important results in probability theory.
- It states that, under certain conditions, the sum of a large number of random variables is approximately normal.
- Here, we state a version of the CLT that applies to i.i.d. random variables.

The Central Limit Theorem (CLT)

Let X_1,X_2,...,X_n be i.i.d. random variables with expected value $EX_i = \mu < \infty$ and variance $0 < \mathrm{Var}(X_i) = \sigma^2 < \infty$. Then, the random variable

$$Z_n = \frac{\overline{X} - \mu}{\sigma / \sqrt{n}} = \frac{X_1 + X_2 + \ldots + X_n - n\mu}{\sqrt{n}\sigma}$$

converges in distribution to the standard normal random variable as n goes to infinity, that is

$$\lim_{n \to \infty} P(Z_n \leq x) = \Phi(x), \qquad \text{for all } x \in \mathbb{R},$$

where $\Phi(x)$ is the standard normal CDF.

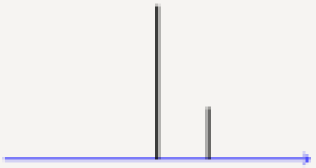
Assumptions:

- $X_1, X_2 \ldots$ are iid Bernoulli(p).
- $Z_n = \dfrac{X_1 + X_2 + \ldots + X_n - np}{\sqrt{np(1-p)}}$.

We choose $p = \frac{1}{3}$.

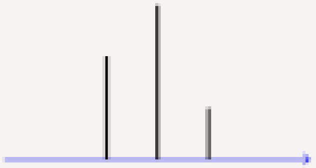
$$Z_1 = \frac{X_1 - p}{\sqrt{p(1-p)}}$$

PMF of Z_1



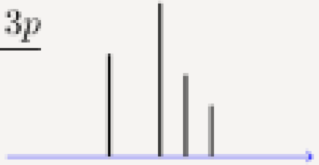
$$Z_2 = \frac{X_1 + X_2 - 2p}{\sqrt{2p(1-p)}}$$

PMF of Z_2



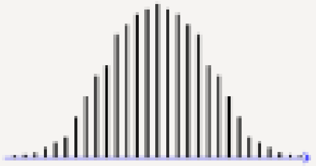
$$Z_3 = \frac{X_1 + X_2 + X_3 - 3p}{\sqrt{3p(1-p)}}$$

PMF of Z_3



$$Z_{30} = \frac{\sum_{i=1}^{30} X_i - 30p}{\sqrt{30p(1-p)}}$$

PMF of Z_{30}



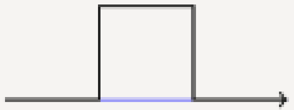
Z_n is the normalized sum of n independent $Bernoulli(p)$ random variables.
shape of its PMF, $P_{Z_n}(z)$, resembles the normal curve as n increases.

Assumptions:

- $X_1, X_2 \ldots$ are iid Uniform(0,1).
- $Z_n = \dfrac{X_1 + X_2 + \ldots + X_n - \frac{n}{2}}{\sqrt{\frac{n}{12}}}$.

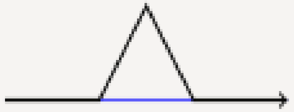
$$Z_1 = \frac{X_1 - \frac{1}{2}}{\sqrt{\frac{1}{12}}}$$

PDF of Z_1



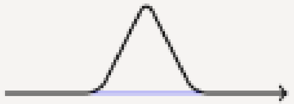
$$Z_2 = \frac{X_1 + X_2 - 1}{\sqrt{\frac{2}{12}}}$$

PDF of Z_2



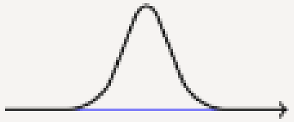
$$Z_3 = \frac{X_1 + X_2 + X_3 - \frac{3}{2}}{\sqrt{\frac{3}{12}}}$$

PDF of Z_3



$$Z_{30} = \frac{\sum_{i=1}^{30} X_i - \frac{30}{2}}{\sqrt{\frac{30}{12}}}$$

PDF of Z_{30}



Z_n is the normalized sum of n independent $Uniform(0,1)$ random variables. shape of its PDF, $f_{Z_n}(z)$, gets closer to the normal curve as n increases.

How to Apply The Central Limit Theorem (CLT)

Here are the steps that we need in order to apply the CLT:

1. Write the random variable of interest, Y, as the sum of n i.i.d. random variable X_i's:

$$Y = X_1 + X_2 + \ldots + X_n.$$

2. Find EY and $\mathrm{Var}(Y)$ by noting that

$$EY = n\mu, \qquad \mathrm{Var}(Y) = n\sigma^2,$$

where $\mu = EX_i$ and $\sigma^2 = \mathrm{Var}(X_i)$.

3. According to the CLT, conclude that $\frac{Y-EY}{\sqrt{\mathrm{Var}(Y)}} = \frac{Y-n\mu}{\sqrt{n}\sigma}$ is approximately standard normal; thus, to find $P(y_1 \leq Y \leq y_2)$, we can write

$$\begin{aligned} P(y_1 \leq Y \leq y_2) &= P\left(\frac{y_1 - n\mu}{\sqrt{n}\sigma} \leq \frac{Y - n\mu}{\sqrt{n}\sigma} \leq \frac{y_2 - n\mu}{\sqrt{n}\sigma}\right) \\ &\approx \Phi\left(\frac{y_2 - n\mu}{\sqrt{n}\sigma}\right) - \Phi\left(\frac{y_1 - n\mu}{\sqrt{n}\sigma}\right). \end{aligned}$$

Example

A bank teller serves customers standing in the queue one by one. Suppose that the service time X_i for customer i has mean $EX_i = 2$ (minutes) and $\mathrm{Var}(X_i) = 1$. We assume that service times for different bank customers are independent. Let Y be the total time the bank teller spends serving 50 customers. Find $P(90 < Y < 110)$.

Example

Let $\overline{X}_{25}$ denote the sample mean of 25 random variables where each random variable comes from a distribution with a mean of 15 and variance of 4. Find $P(14.4 \leq \overline{X}_{25} \leq 15.6)$.